

Project Concept Note: SafeBox Web

Project Title and Application Name

Project Title: AI-Powered URL Safety Analyzer **Application Name:** SafeBox Web

Problem Statement / Objective

With the increasing sophistication of phishing attacks and credential harvesting campaigns, users often struggle to determine the safety of a URL before clicking it. Traditional blocklists are reactive and often miss zero-day threats or highly targeted spear-phishing links. The objective of SafeBox Web is to provide a proactive, two-tier analysis system that combines rapid structural heuristics with deep, context-aware Large Language Model (LLM) evaluation to assess the malicious intent of any given URL in real-time.

Target User and Use Case

Target User: General internet users, remote workers, and small business employees who frequently receive links via email, SMS, or messaging platforms and need a quick, reliable way to verify their safety. **Use Case:** A user receives a suspicious email claiming their bank account is locked, containing a link. Instead of clicking the link, the user copies the URL, pastes it into the SafeBox Web dashboard, and clicks "Analyze." The application immediately performs a structural check and then streams a detailed AI-generated report explaining whether the link is safe, highlighting specific red flags like typosquatting or suspicious TLDs.

LLM Model and API Used

Model: OpenAI GPT-4o (or Anthropic Claude 3.5 Sonnet, depending on configuration) **API:** OpenAI Chat Completions API (or Anthropic Messages API)

Key Features of the Application

1. Two-Tier Analysis Engine:

- **Tier 1 (Fast Check):** A rapid, server-side Python function that evaluates the URL structure for common phishing indicators, including typosquatting against major brands, suspicious Top-Level Domains (TLDs), IP-based hostnames, excessive subdomains, and punycode (homograph attacks).

- **Tier 2 (AI Deep Analysis):** Integration with an advanced LLM to evaluate the URL's context, potential credential harvesting intent, and overall safety profile.
- 2. **Real-Time Streaming Responses:** The backend streams the LLM's analysis chunk-by-chunk to the frontend, providing immediate visual feedback and a progressive reading experience.
- 3. **Responsive, Cybersecurity-Themed UI:** A clean, modern, dark-mode-ready dashboard built with Vanilla HTML/CSS/JS, optimized for both desktop and mobile devices.
- 4. **Secure Architecture:** API keys are strictly managed server-side via environment variables (`.env`), ensuring no sensitive credentials are exposed to the client.
- 5. **Containerized Deployment:** The entire full-stack application is packaged into a single Docker container for seamless deployment to AWS.

Expected User Experience and Outcomes

Users will experience a frictionless, intuitive interface. Upon submitting a URL, they will immediately see the results of the Tier 1 fast check. If the URL passes or requires further scrutiny, the Tier 2 AI analysis will begin streaming onto the screen, typing out a comprehensive safety report in real-time. The outcome is increased user confidence and a significant reduction in the likelihood of falling victim to phishing or malicious websites.

Live AWS Application URL

(To be updated upon successful AWS deployment)